# iLearn Practice Exam

11BR · Chapter / Topic 1

## Cambridge AS Level Computer Science 9618

Paper 1  Theory Fundamentals

# Topic 1: Information Representation

**Name:** _____

**Class:** _____

**Date:** _____

**Time allowed:** 1 hour 30 minutes

**TOTAL MARKS:** 75

Marks obtained: ..... / 75

**INSTRUCTIONS**

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

**INFORMATION**

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 **(a)** Complete the following description.

A kibibyte has a ..... prefix. Three kibibytes is the same as ..... bytes.

A megabyte has a ..... prefix. Two terabytes is the same as ..... gigabytes.

[4]

**(b)** Convert the denary number 241 to hexadecimal.

.....

..... [1]

**(c)** State what is meant by an **overflow in binary addition**.

.....

..... [1]

**(d)** Computers use character sets when representing characters in binary.

**(i)** Complete the table by identifying the number of bits each of the character sets allocates to each character.

| Character set | Number of bits |
|---|---|
| ASCII | |
| extended ASCII | |
| Unicode | |

[1]

**(ii)** Explain how the word 'Clock' is represented by a character set.

.....

.....

.....

..... [2]

2 A photograph is stored as a bitmap image.

**(a)** The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes.

Calculate an estimate for the file size of the photograph in megabytes.

Show your working.

Working .....

.....

.....

Answer ..... megabytes

[2]

**(b)** The photograph is compressed before being uploaded to a web server.

**(i)** Give **three** benefits of this photograph being compressed using lossy compression instead of lossless compression.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

**(ii)** Explain how run-length encoding (RLE) will compress the photograph.

.....

.....

.....

..... [2]

**(c)** Identify **two** elements of a bitmap image that can be changed to reduce its file size.

1 .....

.....

2 .....

3 **(a)** Convert the denary integer 558 into 12-bit binary and hexadecimal.

Binary .....

Hexadecimal .....
[2]

**(b)** **(i)** Convert the two's complement binary integer 11100010 into denary.

.....

.....

..... [1]

**(ii)** Write the smallest and the largest two's complement binary integers that can be represented in 8 bits.

Smallest .....

Largest .....
[2]

**(c)** Give **one** application where Binary Coded Decimal (BCD) is used and justify its use.

Application .....

Justification .....

.....
[2]

4 **(a)** Tick (✓) **one** box to identify the correct definition of a pixel in a bitmap image.

- ☐ a colour in an image
- ☐ a row of consecutive identical colours in an image
- ☐ the binary code for one colour in an image
- ☐ the smallest element of an image that can be one colour

[1]

**(b)** Describe the difference between the terms image resolution and screen resolution.

.....

.....

.....

..... [2]

**(c)** Two items in a bitmap image file header are the resolution and colour depth.

Identify **one** other item included in the file header of a bitmap image.

.....

..... [1]

**(d)** The following bitmap image is created. Each letter represents a colour: R is red, W is white, B is blue.

| | | | | | |
|---|---|---|---|---|---|
| R | R | W | W | W | W |
| B | B | R | R | W | W |
| W | W | W | R | R | R |
| W | W | W | W | R | R |
| W | W | R | R | R | R |
| R | R | R | R | R | B |
| B | B | B | B | B | B |

**(i)** Identify the smallest bit depth that can be used for the image shown.

..... [1]

**(ii)** Explain how the image shown can be compressed using lossless compression.

.....

.....

.....

.....

.....

..... [3]

5 An animation is a sequence of bitmap images with a sound recording played at the same time.

**(a)** The sound is recorded with a sampling resolution of 2 bytes.

State what is meant by a sampling resolution of 2 bytes.

.....

..... [1]

**(b)** **(i)** Explain how the data for the bitmapped images are encoded.

.....

.....

.....

.....

.....

..... [3]

**(ii)** The bitmap images have a bit depth of 2 bytes. Each image is 2048 pixels wide by 1536 pixels high. One second of video contains eight individual bitmap images.

There is no sound recording in this part of the video.

Calculate an estimate for the file size of one second of the video in mebibytes (MiB). Show all of your working.

working: .....

.....

.....

.....

answer ..... MiB

[2]

**(c)** The sound and images are compressed independently before being transmitted over the internet to public cloud storage.

**(i)** Explain the different ways that lossless compression can be used to compress the images and the sound file.

image file .....

.....

.....

.....

sound file .....

.....

.....

.....

[4]

**(ii)** State what is meant by a public cloud.

.....

..... [1]

**(iii)** Give **two** benefits to the animation creator of using public cloud storage for the animation.

1 .....

.....

2 .....

.....

[2]

6 **(a)** Draw **one or more** lines to match each denary number with its equivalent unsigned binary, hexadecimal or Binary Coded Decimal (BCD) value(s).

| Denary number | Equivalent unsigned value |
|---|---|
| | Hexadecimal: 121 |
| 3120 | BCD: 0001010110000101010001 |
| 289 | Binary: 110000110000 |
| 22516 | Hexadecimal: 57F4 |
| 15851 | Binary: 11110111101011 |
| | BCD: 0011000100100000 |

[4]

**(b)** Complete the binary addition. Show all of your working.

1101010011111

+1111110011111

[2]

**(c)** Convert the two's complement integer 011010110001 into denary.

.....

..... [1]

**(d)** Tick (✓) **one** box to identify the smallest integer that can be represented by a 16-bit two's complement integer.

- ☐ 0000000000000000
- ☐ 1111111111111111
- ☐ 0111111111111110
- ☐ 1000000000000000

[1]

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7 **(a)** A real-time video of a music concert needs to be streamed to subscribers.

Tick (✓) **one** box to identify the most appropriate type of compression **and** justify your answer.

| Lossy | Lossless |
|---|---|
| | |

Justification .....

.....

.....

.....

.....

..... [3]

**(b)** Explain the impact of changing the sampling resolution on the accuracy of a sound recording.

.....

.....

.....

.....

.....

..... [3]

**(c)** A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working .....

.....

.....

.....

.....

.....

Estimated file size in mebibytes .....

[2]

8 A computer system stores text, images and sound.

**(a)** A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set .....

Description .....

.....

[2]

**(b)** The colour of each pixel in a bitmapped image is represented by 8 bits.

**(i)** State the largest number of different colours that can be represented by 8 bits.

..... [1]

**(ii)** State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

.....

..... [1]

**(iii)** A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

.....

.....

.....

..... [2]

**(c)** **(i)** Explain how an analogue sound wave is converted into digital data.

.....

.....

.....

..... [2]

**(ii)** Describe **one** method of compressing a sound file using lossy compression.

.....

.....

.....

..... [2]

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9 (a) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

Explanation .....

.....

.....

.....

.....

Denary value .....

[3]

**END OF QUESTION PAPER**

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